



CSCE 633-700: Machine Learning

Linear regression

Overview

Linear Regression

- Example
- Representation, Evaluation
- Optimization: Closed form solution via Ordinary Least Squares
- Optimization: Numerical solution via Gradient Descent
 - General gradient descent
 - Gradient descent for linear regression (batch, stochastic, mini-batch)
- Non-linear basis function for regression & Overfitting

[Parts of these slides have been adapted from K. Murphy (Machine Learning: A probabilistic perspective), Dr. Andrew Ng's Machine Learning course at Coursera, and CSCI567 Machine Learning (USC, Drs. Sha & Liu)]